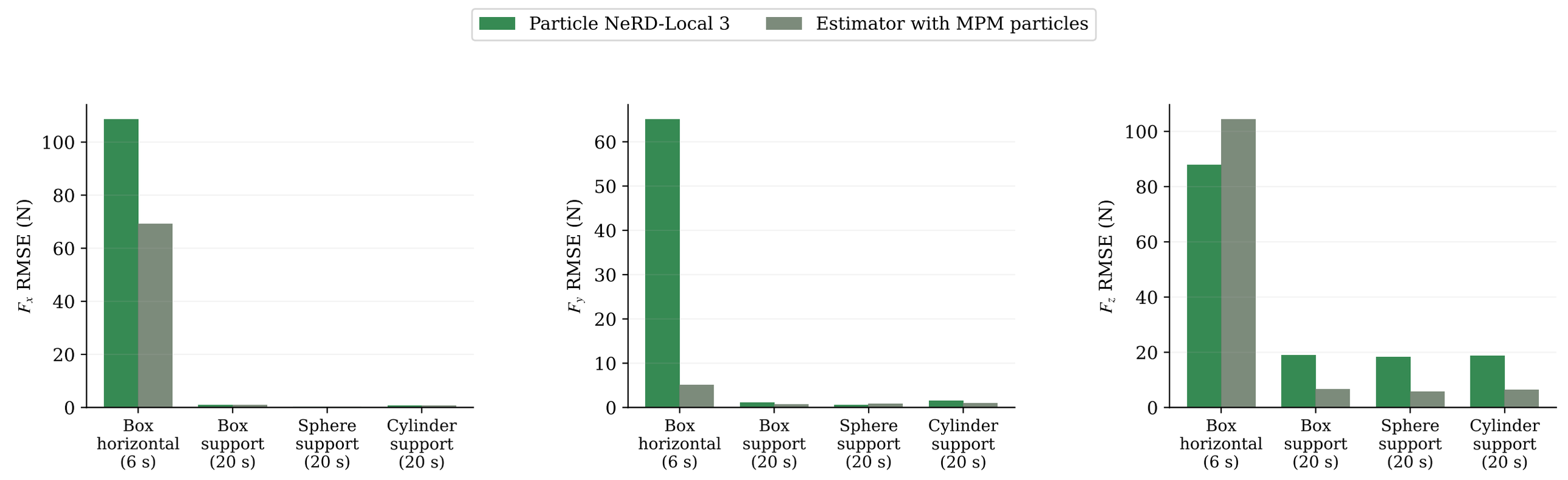

Figure 5: Frozen object-model checkpoint 4,500. Columns show Fx, Fy, Fz. Top row: component RMSE; bottom row: signed mean error (estimate minus native MPM material force). Direct rigid-tray force is excluded. Each case is one raw 200 Hz recording. The gray diagnostic supplies MPM particles to the same estimator. No force reference exists for the 60-test dataset.